

4 章 指数関数と対数関数

1 指数関数

BASIC

■201 (1)

$$\sqrt[3]{6}$$

■201 (2)

$$\pm \sqrt[4]{10}$$

■201 (3)

$$\sqrt[5]{-12}$$

■202 (1)

$$\begin{aligned}\sqrt{2} \times \sqrt{32} \\&= \sqrt{2} \times 4\sqrt{2} \\&= 8\end{aligned}$$

■202 (2)

$$\begin{aligned}(\sqrt[5]{5})^3 (\sqrt[5]{25}) \\&= 5^{\frac{3}{5}} \times 5^{\frac{2}{5}} \\&= 5^1 \\&= 5\end{aligned}$$

■202 (3)

$$\begin{aligned}\sqrt[4]{27} \times \sqrt[4]{3} \\&= 3^{\frac{3}{4}} \times 3^{\frac{1}{4}} \\&= 3^1 \\&= 3\end{aligned}$$

■202 (4)

$$\begin{aligned}\frac{\sqrt[3]{48}}{\sqrt[3]{6}} \\&= 8^{\frac{1}{3}} \\&= 2\end{aligned}$$

■203 (1)

$$\begin{aligned}3^8 \times \left(\frac{1}{9}\right)^3 \\&= 3^8 \times 3^{-6} \\&= 3^2 \\&= 9\end{aligned}$$

■203 (2)

$$\begin{aligned}10^2 \times 5^{-3} \times \left(\frac{1}{4}\right)^{-1} \\&= 2^2 \times 5^2 \times 5^{-3} \times 2^2 \\&= 2^4 \times 5^{-1} \\&= \frac{16}{5}\end{aligned}$$

■203 (3)

$$\begin{aligned}(ab^3)^2 \times (a^{-1}b^2)^{-1} \\&= a^2b^6 \times a^1b^{-2} \\&= a^3b^4\end{aligned}$$

■203 (4)

$$\begin{aligned}\frac{(9ab^{-2})^2}{(3a^{-2}b^2)^3} \\&= 3^4a^2b^{-4} \times 3^{-3}a^6b^{-6} \\&= 3a^8b^{-10} \\&= \frac{3a^8}{b^{10}}\end{aligned}$$

■204 (1)

$$\sqrt{a^5} = a^{\frac{5}{2}}$$

■204 (2)

$$\frac{1}{\sqrt[7]{a^3}} = a^{-\frac{3}{7}}$$

■204 (3)

$$a^{\frac{2}{5}} = \sqrt[5]{a^2}$$

■204 (4)

$$a^{-\frac{1}{2}} = \frac{1}{\sqrt{a}}$$

■205 (1)

$$\left(a^{-\frac{1}{2}}\right)^{\frac{4}{3}} = a^{-\frac{2}{3}}$$

■205 (2)

$$\begin{aligned}\frac{a^{\frac{1}{2}}}{a^{-\frac{1}{4}}} &= a^{\frac{1}{2}} \times a^{\frac{1}{4}} \\ &= a^{\frac{3}{4}}\end{aligned}$$

■205 (3)

$$\begin{aligned}a^{1.2} \times a^{0.4} &= a^{1.6} \\ &= a^{\frac{8}{5}}\end{aligned}$$

■206 (1)

$$\begin{aligned}\sqrt[4]{a^2} \times \sqrt[5]{a^3} \\ &= a^{\frac{2}{4}} \times a^{\frac{3}{5}} \\ &= a^{\frac{5}{10}} \times a^{\frac{6}{10}} \\ &= a^{\frac{11}{10}}\end{aligned}$$

■206 (2)

$$\begin{aligned}\frac{a\sqrt[4]{a}}{\sqrt{a}} &= a^1 \times a^{\frac{1}{4}} \times a^{-\frac{1}{2}} \\ &= a^{\frac{3}{4}}\end{aligned}$$

■206 (3)

$$\begin{aligned}\sqrt[3]{\sqrt[4]{a}} &= \left(a^{\frac{1}{4}}\right)^{\frac{1}{3}} \\ &= a^{\frac{1}{12}}\end{aligned}$$

■207 (1)

$$\begin{aligned}y &= 4^x \\ (0, 1), (1, 4) &\text{を通る増加関数} \\ \text{漸近線 } y &= 0\end{aligned}$$

■207 (2)

$$\begin{aligned}y &= 4^{x-1} \\ y &= 4^x \text{ のグラフを } x \text{ 軸方向に } 1 \\ &\text{平行移動したもの}\end{aligned}$$

■207 (3)

$$\begin{aligned}y &= \left(\frac{1}{4}\right)^x \\ &= 4^{-x} \\ y &= 4^x \text{ のグラフと } y \text{ 軸対称}\end{aligned}$$

■208 (1)

$$\begin{aligned}y &= -4^x \\ y &= 4^x \text{ のグラフと } x \text{ 軸対称}\end{aligned}$$

■208 (2)

$$\begin{aligned}y &= -4^{-x} \\ y &= 4^x \text{ のグラフと原点对称}\end{aligned}$$

■209 (1)

$$\begin{aligned}2^{2x} &= 2^{\sqrt[3]{2}} \\ 2^{2x} &= 2^{1+\frac{1}{3}} \\ 2x &= \frac{4}{3} \\ x &= \frac{2}{3}\end{aligned}$$

■209 (2)

$$\begin{aligned}3^{-x+1} &= \sqrt{243} \\ 3^{-x+1} &= 3^{\frac{5}{2}} \\ -x+1 &= \frac{5}{2} \\ x &= -\frac{3}{2}\end{aligned}$$

■209 (3)

$$\begin{aligned}9^x - 7 \cdot 3^x - 18 &= 0 \\ (3^x)^2 - 7 \cdot 3^x - 18 &= 0 \\ (3^x - 9)(3^x + 2) &= 0 \\ 3^x &= 9, -2 \\ 3^x &= -2 \text{ は解なし} \\ x &= 2\end{aligned}$$

■210 (1)

$$\begin{aligned}8^x &> 16 \\ 2^{3x} &> 2^4 \\ \text{底 } 2 &> 1 \text{ なので } 3x &> 4 \\ x &> \frac{4}{3}\end{aligned}$$

■210 (2)

$$\left(\frac{1}{5}\right)^{x+3} < \frac{1}{25}$$

$$\left(\frac{1}{5}\right)^{x+3} < \left(\frac{1}{5}\right)^2$$

底 $\frac{1}{5} < 1$ なので $x+3 > 2$

$$x > -1$$

CHECK

■211 (1)

$$\begin{aligned}\sqrt[3]{-8} \times \sqrt[5]{-243} \\ &= (-2) \times (-3) \\ &= 6\end{aligned}$$

■211 (2)

$$\begin{aligned}\sqrt[4]{(-2)^4} \times \sqrt[3]{125} \\ &= 2 \times 5 \\ &= 10\end{aligned}$$

■211 (3)

$$\begin{aligned}(\sqrt[3]{2})^5 \times \sqrt[3]{54} \quad (54 = 2 \times 3^3) \\ &= 2^{\frac{5}{3}} \times 2^{\frac{1}{3}} \times 3^{\frac{3}{3}} \\ &= 2^2 \times 3 \\ &= 12\end{aligned}$$

■211 (4)

$$\begin{aligned}\frac{\sqrt[5]{24}}{(\sqrt[5]{18})^3} \\ &= \left(\frac{24}{18^3}\right)^{\frac{1}{5}} \\ &= \left(\frac{2^3 \times 3}{2^3 \times 3^6}\right)^{\frac{1}{5}} \\ &= \frac{1}{3}\end{aligned}$$

■212 (1)

$$a^2 \sqrt{a} = a^2 \times a^{\frac{1}{2}} = a^{\frac{5}{2}}$$

■212 (2)

$$\sqrt{\sqrt[3]{a^2}} = \left(a^{\frac{2}{3}}\right)^{\frac{1}{2}} = a^{\frac{1}{3}}$$

■212 (3)

$$\frac{1}{a^2 \sqrt[4]{a^3}} = a^{-2} \times a^{-\frac{3}{4}} = a^{-\frac{11}{4}}$$

■212 (4)

$$\begin{aligned}\frac{a^2}{\sqrt[10]{\sqrt[5]{a}}} &= a^2 \times \left(a^{\frac{1}{5}}\right)^{-\frac{1}{10}} \\ &= a^2 \times a^{-\frac{1}{50}} = a^{\frac{99}{50}}\end{aligned}$$

■213 (1)

$$a^{\frac{1}{2}} \times a^{\frac{1}{3}} = a^{\frac{5}{6}} = \sqrt[6]{a^5}$$

■213 (2)

$$\begin{aligned}\frac{a^{\frac{5}{4}}}{a^{\frac{3}{2}}} &= a^{\frac{5}{4}} \times a^{-\frac{3}{2}} \\ &= a^{-\frac{1}{4}} = \frac{1}{\sqrt[4]{a}}\end{aligned}$$

■214 (1)

$$\begin{aligned}(2a^2)^{\frac{2}{3}} \times (2a^{-1})^{\frac{1}{3}} \\ &= 2^{\frac{2}{3}} \cdot a^{\frac{4}{3}} \cdot 2^{\frac{1}{3}} \cdot a^{-\frac{1}{3}} \\ &= 2^1 \cdot a^1 \\ &= 2a\end{aligned}$$

■214 (2)

$$\begin{aligned}\left(a^{\frac{1}{2}} + a^{-\frac{1}{2}}\right)^2 \\ &= a^{2 \cdot \frac{1}{2}} + 2 \cdot a^{\frac{1}{2}} \cdot a^{-\frac{1}{2}} + a^{-\frac{1}{2} \cdot 2} \\ &= a + 2 + \frac{1}{a}\end{aligned}$$

■214 (3)

$$\begin{aligned}\frac{\sqrt[3]{a^2} \times \sqrt{a}}{\sqrt[6]{a^5}} \\ &= a^{\frac{2}{3}} \times a^{\frac{1}{2}} \times a^{-\frac{5}{6}} \\ &= a^{\frac{4+3-5}{6}} \\ &= a^{\frac{1}{3}}\end{aligned}$$

■214 (4)

$$\begin{aligned}\left(\frac{\sqrt[4]{a}}{\sqrt[3]{a}}\right)^2 \times \sqrt[6]{a} \\ &= a^{\frac{2}{4}} \times a^{-\frac{2}{3}} \times a^{\frac{1}{6}} \\ &= a^{\frac{6-8+2}{12}} \\ &= 1\end{aligned}$$

■215 (1)

y 方向に $+1$

■215 (2)

$$\begin{aligned}y = 3^x \rightarrow -y = 3^{-x} \\ \therefore \text{原点对称}\end{aligned}$$

■216 (1)

$$\begin{aligned}4^x - 7 \cdot 2^x - 8 &= 0 \\(2^x - 8)(2^x + 1) &= 0 \\2^x &= 8, -1 \\x &= 3\end{aligned}$$

■216 (2)

$$\begin{aligned}2 \left(\frac{1}{4}\right)^x - 9 \left(\frac{1}{2}\right)^x + 4 &= 0 \\2 - 9 \cdot 2^x + 4(2^x)^2 &= 0 \\(4 \cdot 2^x - 1)(2^x - 2) &= 0 \\2^x &= \frac{1}{4}, 2 \\x &= 1, -2\end{aligned}$$

■217 (1)

$$\begin{aligned}3^{2x-1} &< \frac{1}{\sqrt[3]{81}} \\3^{2x-1} &< 3^{-\frac{4}{3}} \\ \text{底 } 3 > 1 \text{ なので} \\2x - 1 &< -\frac{4}{3} \\2x &< -\frac{1}{3} \\x &< -\frac{1}{6}\end{aligned}$$

■217 (2)

$$\begin{aligned}\left(\frac{1}{2}\right)^x &< \sqrt[3]{32} \\2^{-x} &< 2^{\frac{5}{3}} \\ \text{底 } 2 > 1 \text{ なので} \\-x &< \frac{5}{3} \\x &> -\frac{5}{3}\end{aligned}$$

STEP UP

■218 (1)

$$\begin{aligned}\sqrt[5]{81} &= 3^{\frac{4}{5}} \\ \frac{1}{\sqrt[3]{9}} &= 3^{-\frac{2}{3}} \\ 3 \cdot \sqrt[4]{3} &= 3^{\frac{5}{4}} \\ 1 &= 3^0 \\ -\frac{2}{3} < 0 < \frac{4}{5} < \frac{5}{4} \\ \text{底 } 3 > 1 \text{ なので} \\ \frac{1}{\sqrt[3]{9}} < 1 < \sqrt[5]{81} < 3 \cdot \sqrt[4]{3}\end{aligned}$$

■218 (2)

$$\begin{aligned}\sqrt[5]{0.7} &= 0.7^{\frac{1}{5}} \\ \frac{1}{0.49} &= 0.7^{-2} \\ \frac{1}{\sqrt[4]{0.7}} &= 0.7^{-\frac{1}{4}} \\ \left(\sqrt[3]{0.7}\right)^4 &= 0.7^{\frac{4}{3}} \\ -2 < -\frac{1}{4} < \frac{1}{5} < \frac{4}{3} \\ \text{底 } 0.7 < 1 \text{ なので} \\ \left(\sqrt[3]{0.7}\right)^4 &< \sqrt[5]{0.7} \\ &< \frac{1}{\sqrt[4]{0.7}} < \frac{1}{0.49}\end{aligned}$$

■219 (1)

$$\begin{aligned}0.5^{2x+1} &> 0.125^{1-x} \\ \left(\frac{1}{2}\right)^{2x+1} &> \left(\frac{1}{8}\right)^{1-x} \\ 2^{-(2x+1)} &> 2^{-3(1-x)} \\ \text{底 } 2 > 1 \text{ なので} \\ -2x - 1 &> -3 + 3x \\ -5x &> -2 \\ x &< \frac{2}{5}\end{aligned}$$

■219 (2)

$$\begin{aligned}\sqrt[4]{a^{3x}} &> \frac{1}{a^{x+1}} \\ a^{\frac{3}{4}x} &> a^{-(x+1)}\end{aligned}$$

i) 底 $a > 1$ のとき

$$\begin{aligned}\frac{3}{4}x &> -x - 1 \\ \frac{7}{4}x &> -1 \\ x &> -\frac{4}{7}\end{aligned}$$

ii) 底 $0 < a < 1$ のとき

$$\begin{aligned}\frac{3}{4}x &< -x - 1 \\ x &< -\frac{4}{7}\end{aligned}$$

■220 (1)

$$\begin{aligned}2^{2x+3} - 17 \cdot 2^x + 2 &> 0 \\ 8 \cdot (2^x)^2 - 17 \cdot 2^x + 2 &> 0 \\ (8 \cdot 2^x - 1)(2^x - 2) &> 0 \\ 2^x &< \frac{1}{8}, 2 < 2^x \\ \text{底 } 2 > 1 \text{ なので} \\ x &< -3, 1 < x\end{aligned}$$

■220 (2)

$$\begin{aligned}3^{2x} - 8 \cdot 3^x - 9 &< 0 \\ (3^x - 9)(3^x + 1) &< 0 \\ -1 &< 3^x < 9 \\ \text{底 } 3 > 1 \text{ なので} \\ x &< 2\end{aligned}$$

■221

$$\begin{aligned}3^x + 3^{-x} &= 5 \\ \text{両辺 } 3 \text{ 乗して} \\ 3^{3x} + 3 \cdot 3^{2x} \cdot 3^{-x} \\ &+ 3 \cdot 3^x \cdot 3^{-2x} + 3^{-3x} = 125 \\ 27^x + 27^{-x} + 3(3^x + 3^{-x}) &= 125 \\ 27^x + 27^{-x} + 15 &= 125 \\ 27^x + 27^{-x} &= 110\end{aligned}$$

■222

$$\begin{aligned}
 & \frac{a^{4x} - a^{-4x}}{a^x - a^{-x}} \\
 &= \frac{(a^{2x} + a^{-2x})(a^{2x} - a^{-2x})}{a^x - a^{-x}} \\
 &= \frac{(a^{2x} + a^{-2x})(a^x + a^{-x})(a^x - a^{-x})}{a^x - a^{-x}} \\
 &= \left(6 + \frac{1}{6}\right) \left(\sqrt{6} + \frac{1}{\sqrt{6}}\right) \quad (\because a > 0) \\
 &= \frac{37}{6} \cdot \frac{7}{\sqrt{6}} \\
 &= \frac{259}{36} \sqrt{6}
 \end{aligned}$$

■223 (1)

$$\begin{aligned}
 y &= 9^x - 6 \cdot 3^x + 5 \quad (0 \leq x \leq 2) \\
 &= (3^x - 3)^2 - 4 \\
 1 &\leq 3^x \leq 9 \text{ なので} \\
 3^x &= 3 \text{ すなわち } x = 1 \text{ で最小値 } -4 \\
 3^x &= 9 \text{ すなわち } x = 2 \text{ で最大値 } 32 \\
 &\text{をとる}
 \end{aligned}$$

■223 (2)

$$\begin{aligned}
 y &= (3^x - 3)^2 - 4 \quad (-1 \leq x \leq 0) \\
 \frac{1}{3} &\leq 3^x \leq 1 \text{ なので} \\
 3^x &= 1 \text{ すなわち } x = 0 \text{ で最小値 } 0 \\
 3^x &= \frac{1}{3} \text{ すなわち } x = -1 \text{ で} \\
 &\text{最大値 } \frac{64}{9} - 4 = \frac{28}{9} \text{ をとる}
 \end{aligned}$$

■224 (1)

$$\begin{aligned}
 X^2 &= (2^x + 2^{-x})^2 \\
 &= 4^x + 2 + 4^{-x} \\
 \therefore 4^x + 4^{-x} &= X^2 - 2 \\
 f(x) &= (X^2 - 2) - 2X + 1 \\
 &= X^2 - 2X - 1 \\
 &= (X - 1)^2 - 2
 \end{aligned}$$

■224 (2)

$$\begin{aligned}
 &2^x > 0, 2^{-x} > 0 \text{ なので} \\
 &\text{相加平均と相乗平均の関係より} \\
 X &\geq 2\sqrt{2^x \cdot 2^{-x}} \\
 &= 2 \\
 &\text{等号成立は } 2^x = 2^{-x} \\
 2^{2x} &= 1 \\
 &\text{すなわち } x = 0 \text{ のとき}
 \end{aligned}$$

■224 (3)

$$\begin{aligned}
 f(x) &= (X - 1)^2 - 2 \quad (X \geq 2) \\
 X &= 2 \text{ すなわち } x = 0 \text{ で最小値 } -1 \\
 &\text{をとる}
 \end{aligned}$$